

Kochenov Equation: Geometric Solar-System Scaling Relation

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Abstract

A dimensionless geometric ratio constructed from the mean radii of the Sun, Moon, and Earth:

$$\Gamma = \frac{R_{\odot} \times R_{\text{Moon}}}{R_{\oplus}^2} \approx 29.7788$$

directly reproduces Earth's orbital speed when multiplied by the unit velocity scale $V_0 = 1$ km/s:

$$v_{\oplus} = \Gamma \times V_0 \approx 29.7788 \text{ km/s.}$$

The observed mean orbital speed is ≈ 29.7847 km/s (deviation 0.02%). From this relation, the solar gravitational parameter follows without using G or any masses:

$$GM_{\odot} = r_{\text{AU}} \times V_0^2 \times \Gamma^2 \approx 1.32660 \times 10^{20} \text{ m}^3/\text{s}^2.$$

IAU/CODATA reference: $1.32712 \times 10^{20} \text{ m}^3/\text{s}^2$ (deviation 0.04%). No free parameters. The Moon's radius is essential — without it the relation does not close.

1 The Geometric Ratio Γ

Using IAU mean radii:

$$R_{\odot} = 695700 \text{ km}, \quad R_{\text{Moon}} = 1737.4 \text{ km}, \quad R_{\oplus} = 6371 \text{ km.}$$

$$\Gamma = \frac{R_{\odot} \times R_{\text{Moon}}}{R_{\oplus}^2} = \frac{695700 \times 1737.4}{6371^2} \approx 29.7788.$$

Γ is dimensionless — independent of the choice of length units.

2 Earth's Orbital Speed

Define the unit velocity scale $V_0 = 1$ km/s. Then:

$$v_{\oplus}^{(\text{geom})} = \Gamma \times V_0 \approx 29.7788 \text{ km/s.}$$

Quantity	Value	Gap
$v_{\oplus}^{(\text{geom})}$	29.7788 km/s	—
$v_{\oplus}^{(\text{observed})}$	29.7847 km/s	−0.02%

No gravitational constant G , no masses — only geometry and a unit scale.

3 Solar Gravitational Parameter

From classical mechanics $v_{\oplus}^2 = GM_{\odot}/r_{\text{AU}}$, with $r_{\text{AU}} = 149597870.7$ km. Substituting the geometric $v_{\oplus}$:

$$GM_{\odot}^{(\text{geom})} = r_{\text{AU}} \times V_0^2 \times \Gamma^2 \approx 1.32660 \times 10^{20} \text{ m}^3/\text{s}^2.$$

Source	$GM_{\odot}$ (m^3/s^2)	Gap
Geometric (this work)	1.32660×10^{20}	—
IAU/CODATA	1.32712×10^{20}	−0.04%

Again, no G and no masses — only geometry, distance, and V_0 .

4 G as a Derived Quantity

The Newtonian constant G appears only if one insists on expressing $GM_{\odot}$ in terms of $M_{\odot}$:

$$G^{(\text{geom})} = \frac{GM_{\odot}^{(\text{geom})}}{M_{\odot}} \approx 6.67144 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}.$$

CODATA 2022: 6.67430×10^{-11} (deviation −0.043%).

In the primary relations $v_{\oplus} = \Gamma V_0$ and $GM_{\odot} = r_{\text{AU}} V_0^2 \Gamma^2$, G is **not needed at all**.

5 Key Features

- **No masses** in the primary relations — only radii, distance, and a unit velocity scale.
- **No free parameters** — everything is fixed by the geometry of the Sun–Earth–Moon system.
- **The Moon’s radius is essential** — without it the system does not close.
- G is a **derived artifact** — it appears only if one insists on separating $GM_{\odot}$.

6 Kochenov Equation (Final Form)

$$v_{\oplus} = \Gamma \times V_0$$

where

$$\Gamma = \frac{R_{\odot} \times R_{\text{Moon}}}{R_{\oplus}^2}, \quad V_0 = 1 \text{ km/s}.$$

Or equivalently, in explicit numerical form:

$$v_{\oplus} \text{ (km/s)} = \frac{R_{\odot} \times R_{\text{Moon}}}{R_{\oplus}^2}$$

7 Conclusion

The dimensionless ratio

$$\Gamma = \frac{R_{\odot} \times R_{\text{Moon}}}{R_{\oplus}^2} \approx 29.7788$$

determines Earth’s orbital speed ($v_{\oplus} = \Gamma \times 1$ km/s, deviation 0.02%) and the solar gravitational parameter ($GM_{\odot} = r_{\text{AU}} \Gamma^2$, deviation 0.04%) without using G or any masses.

The relations are empirical geometric correlations presented as observations. No dynamical model is claimed.

I am almost joking. But the formula works.

— Boris V. Kochenov, May 21, 2026

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